

Question			Marking details	Marks available				Maths	Prac
				AO1	AO2	AO3	Total		
5	(a)	(i)	$\lambda \geq 25 \text{ mm}$ / gap / w . [Accept $>$, not $=$]	1			1		1
		(ii)	Intensity increased straight in front of gap (or equivalent) [or more total power – accept total intensity – passes through gap] (1) Intensity reduced at large angles to normal / 'at the sides' [accept: waves don't diffract as much] (1)			2	2		2
	(b)		$S_1P = 250 \text{ mm}$ (1) $S_2P = 264 \text{ mm}$ (1) Path difference = 14 mm or by implication ecf on S_1P and S_2P (1) $\lambda = 28 \text{ mm}$ (1) NB Use of Young slits formula $\rightarrow 0$ marks	1	1 1 1		4	2	
			Question 5 total	2	3	2	7	2	3